



Red Hat OpenShift AI Self-Managed 3.4

Managing resources

Manage administration tasks from the OpenShift AI dashboard

Red Hat OpenShift AI Self-Managed 3.4 Managing resources

Manage administration tasks from the OpenShift AI dashboard

Legal Notice

Copyright © Red Hat.

Except as otherwise noted below, the text of and illustrations in this documentation are licensed by Red Hat under the Creative Commons Attribution–Share Alike 3.0 Unported license . If you distribute this document or an adaptation of it, you must provide the URL for the original version.

Red Hat, as the licensor of this document, waives the right to enforce, and agrees not to assert, Section 4d of CC-BY-SA to the fullest extent permitted by applicable law.

Red Hat, the Red Hat logo, JBoss, Hibernate, and RHCE are trademarks or registered trademarks of Red Hat, LLC. or its subsidiaries in the United States and other countries.

Linux[®] is the registered trademark of Linus Torvalds in the United States and other countries.

XFS is a trademark or registered trademark of Hewlett Packard Enterprise Development LP or its subsidiaries in the United States and other countries.

The OpenStack[®] Word Mark and OpenStack logo are trademarks or registered trademarks of the Linux Foundation, used under license.

All other trademarks are the property of their respective owners.

Abstract

As an OpenShift AI administrator, manage custom workbench images, cluster PVC size, user groups, and Jupyter notebook servers.

Table of Contents

PREFACE	4
CHAPTER 1. SELECTING OPENSIFT AI ADMINISTRATOR AND USER GROUPS	5
CHAPTER 2. CUSTOMIZING THE DASHBOARD	6
2.1. EDITING THE DASHBOARD CONFIGURATION	6
2.2. DASHBOARD CONFIGURATION OPTIONS	7
CHAPTER 3. IMPORTING A CUSTOM WORKBENCH IMAGE	13
CHAPTER 4. INTRODUCING THE KUBERNETES GATEWAY API FOR CUSTOM IMAGE MIGRATION	15
4.1. UNDERSTANDING PATH-BASED ROUTING	15
4.2. MIGRATING YOUR CUSTOM WORKBENCH IMAGES TO THE KUBERNETES GATEWAY API	15
4.3. MIGRATING YOUR CUSTOM WORKBENCH IMAGES TO THE KUBERNETES GATEWAY API USING NGINX	17
CHAPTER 5. MANAGING CLUSTER PVC SIZE	19
5.1. CONFIGURING THE DEFAULT PVC SIZE FOR YOUR CLUSTER	19
5.2. RESTORING THE DEFAULT PVC SIZE FOR YOUR CLUSTER	19
CHAPTER 6. MANAGING CONNECTION TYPES	21
6.1. VIEWING CONNECTION TYPES	21
6.2. CREATING A CONNECTION TYPE	21
6.3. DUPLICATING A CONNECTION TYPE	23
6.4. EDITING A CONNECTION TYPE	23
6.5. ENABLING A CONNECTION TYPE	24
6.6. DELETING A CONNECTION TYPE	25
CHAPTER 7. MANAGING STORAGE CLASSES	26
7.1. ABOUT PERSISTENT STORAGE	26
7.1.1. Storage classes in OpenShift AI	26
7.1.2. Access modes	26
7.1.2.1. Using shared storage (RWX)	27
7.2. CONFIGURING STORAGE CLASS SETTINGS	27
7.3. CONFIGURING THE DEFAULT STORAGE CLASS FOR YOUR CLUSTER	29
7.4. OVERVIEW OF OBJECT STORAGE ENDPOINTS	29
7.4.1. MinIO (On-Cluster)	29
7.4.2. Amazon S3	30
7.4.3. Other S3-Compatible Object Stores	30
7.4.4. Verification and Troubleshooting	31
CHAPTER 8. MANAGING BASIC WORKBENCHES	32
8.1. ACCESSING THE ADMINISTRATION INTERFACE FOR BASIC WORKBENCHES	32
8.2. STARTING BASIC WORKBENCHES OWNED BY OTHER USERS	32
8.3. ACCESSING BASIC WORKBENCHES OWNED BY OTHER USERS	33
8.4. STOPPING BASIC WORKBENCHES OWNED BY OTHER USERS	33
8.5. STOPPING IDLE WORKBENCHES	34
8.6. ADDING WORKBENCH POD TOLERATIONS	35
8.7. TROUBLESHOOTING COMMON PROBLEMS IN WORKBENCHES FOR ADMINISTRATORS	36
8.7.1. A user receives a 404: Page not found error when logging in to Jupyter	36
8.7.2. A user's workbench does not start	37
8.7.3. The user receives a database or disk is full error or a no space left on device error when they run notebook cells	38

CHAPTER 9. MANAGING THE COLLECTION OF USAGE DATA **39**

9.1. USAGE DATA COLLECTION NOTICE FOR OPENSIFT AI 39

9.2. ENABLING USAGE DATA COLLECTION 39

9.3. DISABLING USAGE DATA COLLECTION 40

PREFACE

As an OpenShift AI administrator, you can manage the following resources:

- OpenShift AI admin and user groups
- Dashboard customization
- Custom workbench images
- Cluster PVC size
- Connection types
- Cluster storage classes

You can also specify whether to allow Red Hat to collect data about OpenShift AI usage in your cluster.

CHAPTER 1. SELECTING OPENSIFT AI ADMINISTRATOR AND USER GROUPS

By default, all users authenticated in OpenShift can access OpenShift AI.

Also by default, users with **cluster-admin** permissions are OpenShift AI administrators. A **cluster admin** is a superuser that can perform any action in any project in the OpenShift cluster. When bound to a user with a local binding, they have full control over quota and every action on every resource in the project.

After a **cluster admin** user defines additional administrator and user groups in OpenShift, you can add those groups to OpenShift AI by selecting them in the OpenShift AI dashboard.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The groups that you want to select as administrator and user groups for OpenShift AI already exist in OpenShift. For more information, see [Managing users and groups](#).

Procedure

1. From the OpenShift AI dashboard, click **Settings → User management**.
2. Select your OpenShift AI administrator groups: Under **Red Hat OpenShift AI administrator groups**, click the text box and select an OpenShift group. Repeat this process to define multiple administrator groups.
3. Select your OpenShift AI user groups: Under **Red Hat OpenShift AI user groups**, click the text box and select an OpenShift group. Repeat this process to define multiple user groups.



IMPORTANT

The **system:authenticated** setting allows all users authenticated in OpenShift to access OpenShift AI.

4. Click **Save changes**.

Verification

- Administrator users can successfully log in to OpenShift AI and have access to the **Settings** navigation menu.
- Non-administrator users can successfully log in to OpenShift AI. They can also access and use individual components, such as projects and workbenches.

CHAPTER 2. CUSTOMIZING THE DASHBOARD

The OpenShift AI dashboard provides features that are designed to work for most scenarios. These features are configured in the **Odhdashboardconfig** custom resource (CR).

To see a description of the options in the OpenShift AI dashboard configuration, see [Dashboard configuration options](#).

As an OpenShift AI administrator, you can customize the interface of the dashboard. For example, you can show or hide some of the dashboard navigation menu items. To change the default settings of the dashboard, edit the **Odhdashboardconfig** CR as described in [Editing the dashboard configuration](#).

2.1. EDITING THE DASHBOARD CONFIGURATION

As an OpenShift AI administrator, you can customize the interface of the dashboard by editing the dashboard configuration.

Prerequisites

- You have OpenShift AI administrator privileges.

Procedure

1. Log in to the OpenShift console as a user with OpenShift AI administrator privileges.
2. In the **Administrator** perspective, click **Home** → **API Explorer**.
3. In the search bar, enter **Odhdashboardconfig** to filter by kind.
4. Click the **Odhdashboardconfig** custom resource (CR) to open the resource details page.
5. From the **Project** list, select the OpenShift AI application namespace; the default is **redhat-ods-applications**.
6. Click the **Instances** tab.
7. Click the **odh-dashboard-config** instance to open the details page.
8. Click the **YAML** tab.
9. Edit the values of the options that you want to change.
For example, to show or hide a menu item in the dashboard navigation menu, update the **spec.dashboardconfig** section to edit the relevant dashboard configuration option.



NOTE

If a dashboard configuration option is not included in the **OdhdashboardConfig** CR, the default value is used.

To change the default behavior for such options, edit the **OdhdashboardConfig** CR to add the missing entry to the **spec.dashboardConfig** section, and set the preferred value:

- To show the feature, set the value to **false**
- To hide the feature, set the value to **true**

Example

By default, the **Observe & monitor → Workload metrics** menu item is shown in the dashboard navigation menu. To hide this menu item, set the **disableDistributedWorkloads** value to **true**, as follows:

```
disableDistributedWorkloads: true
```

For more information about dashboard configuration options and their default values, see [Dashboard configuration options](#).

10. Click **Save** to apply your changes and then click **Reload** to synchronize your changes to the cluster.

Verification

- Log in to OpenShift AI and verify that your dashboard configurations apply.

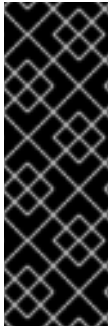
2.2. DASHBOARD CONFIGURATION OPTIONS

The OpenShift AI dashboard includes a set of core features enabled by default that are designed to work for most scenarios. OpenShift AI administrators can configure the OpenShift AI dashboard from the **OdhdashboardConfig** custom resource (CR) in OpenShift.

If a dashboard configuration option is not included in the **OdhdashboardConfig** CR, the default value is used. To change the default behavior for such options, edit the **OdhdashboardConfig** CR to add the missing entry to the **spec.dashboardConfig** section, and set the preferred value:

- To show the feature, set the value to **false**
- To hide the feature, set the value to **true**

For more information about setting dashboard configuration options, see [Editing the dashboard configuration](#).



IMPORTANT

Features denoted with **(Technology Preview)** in this table are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using Technology Preview features in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process. For more information about the support scope of Red Hat Technology Preview features, see [Technology Preview Features Support Scope](#).

Table 2.1. Dashboard feature configuration options

Feature configuration option	Default	Description
spec.dashboardConfig.disableAdminConnectionsTypes	false	Shows the Settings → Environment setup → Connection types menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableBYONImageStream	false	Shows the Settings → Environment setup → Workbench images menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableClusterManager	false	Shows the Settings → Cluster settings menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableCustomServingRuntimes	false	Shows the Settings → Model resources and operations → Serving runtimes menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableDistributedWorkloads	false	Shows the Workload metrics menu item under Observe & monitor in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableFeatureStore	false	Shows the Feature store menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableFineTuning	No longer used	Hides the Models → Model customization menu item in the dashboard navigation menu and the button for registered model versions. To show these items, set the value to false .
spec.dashboardConfig.disableKueue	true	Hides Kueue-related options in the dashboard. Set the value to false to enable Kueue in the dashboard, which allows users to select Kueue-enabled hardware profiles. When false , new projects created from the dashboard are automatically configured for Kueue management with the required label and a default local queue.

Feature configuration option	Default	Description
spec.dashboardConfig.disableLMEval (Technology Preview)	true	Hides the Model → Model evaluation runs menu item in the dashboard navigation menu. To show these items, set the value to false . Model evaluation is a Technology Preview feature in this OpenShift AI release.
spec.dashboardConfig.disableHome	false	Shows the Home menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableInfo	false	On the Applications → Explore page, when a user clicks on an application tile, an information panel opens with more details about the application. To disable the information panel for all applications on the Applications → Explore page, set the value to true .
spec.dashboardConfig.disableISVBadges	false	Shows the label on a tile that indicates whether the application is Red Hat-managed , Partner managed , or Self-managed . To hide these labels, set the value to true .
spec.dashboardConfig.disableKServe	false	Enables the ability to select KServe as a model-serving platform. To disable this ability, set the value to true .
spec.dashboardConfig.disableKServeAuth	false	Enables the ability to use authentication with KServe. To disable this ability, set the value to true .
spec.dashboardConfig.disableKServeMetrics	false	Enables the ability to view KServe metrics. To disable this ability, set the value to true .
spec.dashboardConfig.disableKServeRaw	false	On the Settings → Cluster settings page, in the Model serving platform section, shows the Default deployment mode list.
spec.dashboardConfig.disableLLMd	false	Controls the state of the Enable distributed inference with llm-d switch on the Settings → General settings page. If set to true , the switch is set to Off and the Distributed inference with llm-d option is hidden in the model serving deployment wizard.
spec.dashboardConfig.disableModelCatalog	false	Shows the AI hub → Catalog menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableModelRegistry	false	Shows the AI hub → Registry menu item and the Settings → Model resources and operations → AI registry settings menu item in the dashboard navigation menu. To hide these menu items, set the value to true .

Feature configuration option	Default	Description
spec.dashboardConfig.disableModelRegistrySecureDB	false	Shows the Add CA certificate to secure database connection section in the Create model registry dialog and the Edit model registry dialog. To hide this section, set the value to true .
spec.dashboardConfig.disableModelServing	false	Shows the AI hub → Deployments menu item in the dashboard navigation menu, and the Deployments tab in the Projects menu. To hide these items, set the value to true .
spec.dashboardConfig.disableNIMModelServing	false	Enables the ability to select NVIDIA NIM as a model-serving platform. To disable this ability, set the value to true .
spec.dashboardConfig.disablePerformanceMetrics	false	Shows the Endpoint Performance tab on the Deployments page. To hide this tab, set the value to true .
spec.dashboardConfig.disablePipelines	false	Shows the Develop & train menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.disableProjects	false	Shows the Projects menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.projectRBAC (Technology Preview)	true	Updates the standard Permissions tab on the Projects page to an enhanced version that provides role-based access control (RBAC). To update this tab, set the value to true .
spec.dashboardConfig.disableProjectScoped	false	Distinguishes between global items and project-scoped items (if project-scoped items exist) in the OpenShift AI web console. This option applies to workbench images, hardware profiles, and model-serving runtimes for KServe. To disable this functionality, set the value to true .
spec.dashboardConfig.disableProjectSharing	false	Allows users to share access to their projects with other users. To prevent users from sharing projects, set the value to true .
spec.dashboardConfig.disableServingRuntimeParams	false	Shows the Configuration parameters section in the Deploy model dialog and the Edit model dialog when using the model serving platform. To hide this section, set the value to true .
spec.dashboardConfig.disableStorageClasses	false	Shows the Settings → Cluster settings → Storage classes menu item in the dashboard navigation menu. To hide this menu item, set the value to true .

Feature configuration option	Default	Description
spec.dashboardConfig.disableSupport	false	Shows the Support menu item when a user clicks the Help icon in the dashboard toolbar. To hide this menu item, set the value to true .
spec.dashboardConfig.disableTracking	false	Allows Red Hat to collect data about OpenShift AI usage in your cluster. To disable data collection, set the value to true . You can also set this option in the OpenShift AI dashboard interface from the Settings → Cluster settings navigation menu.
spec.dashboardConfig.disableTrustyBiasMetrics	false	Shows the Model Bias tab on the Models page. To hide this tab, set the value to true .
spec.dashboardConfig.disableUserManagement	false	Shows the Settings → User management menu item in the dashboard navigation menu. To hide this menu item, set the value to true .
spec.dashboardConfig.enablement	true	Enables OpenShift AI administrators to add applications to the OpenShift AI dashboard Applications → Enabled page. To disable this ability, set the value to false .
spec.dashboardConfig.genAiStudio (Technology Preview)	false	Shows or hides the Gen AI studio menu item in the dashboard navigation menu. To show this menu item, set the value to true .
spec.dashboardConfig.mlflow (Developer Preview)	false	Shows or hides the MLflow tile on the Applications → Explore page. To show this tile, set the value to true .
spec.dashboardConfig.modelAsService (Technology Preview)	false	Shows or hides the Model as a service tab on the AI assets endpoint page. To show this tab, set the value to true .
spec.dashboardConfig.observabilityDashboard (Technology Preview)	false	Shows or hides the Observe & monitor → Dashboard menu item in the dashboard navigation menu. To show this tab, set the value to true .
spec.dashboardConfig.trainingJobs (Technology Preview)	true	Shows or hides the Develop and train → Training jobs menu item in the dashboard navigation menu. To show this menu item, set the value to true .

Feature configuration option	Default	Description
spec.groupsConfig	No longer used	Read-only. To configure access to the OpenShift AI dashboard, use the spec.adminGroups and spec.allowedGroups options in the OpenShift Auth resource in the services.platform.opendatahub.io API group.
spec.modelServerSizes	Small, Medium, Large	Allows you to customize names and resources for model servers.
spec.notebookController.enabled	true	Shows the Start basic workbench tile in the Applications section, and the Start basic workbench button on the Projects page. To hide these items, set the value to false .
spec.notebookSizes	Small, Medium, Large, X Large	Allows you to customize names and resources for workbenches. The Kubernetes-style sizes are shown in the drop-down menu that is displayed when launching a workbench with the Notebook Controller. Note: These sizes must follow conventions. For example, requests must be smaller than limits.
spec.templateOrder	[]	Specifies the order of custom Serving Runtime templates. When the user creates a new template, it is added to this list.

CHAPTER 3. IMPORTING A CUSTOM WORKBENCH IMAGE

In addition to workbench images provided and supported by Red Hat and independent software vendors (ISVs), you can import custom workbench images that cater to your project's specific requirements.

You must import it so that your OpenShift AI users (data scientists) can access it when they create a project workbench.

Red Hat supports adding custom workbench images to your deployment of OpenShift AI, ensuring that they are available for selection when creating a workbench. However, Red Hat does not support the contents of your custom workbench image. That is, if your custom workbench image is available for selection during workbench creation, but does not create a usable workbench, Red Hat does not provide support to fix your custom workbench image.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- Your custom image exists in an image registry that is accessible to OpenShift AI.
- The **Settings** → **Environment setup** → **Workbench images** dashboard navigation menu item is enabled, as described in [Enabling custom workbench images in OpenShift AI](#).
- If you want to associate an accelerator with the custom image that you want to import, you know the accelerator's identifier – the unique string that identifies the hardware accelerator. You must also have enabled GPU support in OpenShift AI. This includes installing the Node Feature Discovery Operator and NVIDIA GPU Operator. For more information, see [Installing the Node Feature Discovery Operator](#) and [Enabling NVIDIA GPUs](#).

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Environment setup** → **Workbench images**. The **Workbench images** page opens. Previously imported images are displayed. To enable or disable a previously imported image, on the row containing the relevant image, click the toggle in the **Enable** column.
2. Optional: If you want to associate an accelerator and you have not already created an accelerator profile or a hardware profile, click **Create profile** on the row containing the image and complete the relevant fields. If the image does not contain an accelerator identifier, you must manually configure one before creating an associated accelerator profile or a hardware profile.
3. Click **Import new image**. Alternatively, if no previously imported images were found, click **Import image**. The **Import workbench image** dialog opens.
4. In the **Image location** field, enter the URL of the repository containing the image. For example: **quay.io/my-repo/my-image:tag**, **quay.io/my-repo/my-image@sha256:xxxxxxxxxxxxxx**, or **docker.io/my-repo/my-image:tag**.
5. In the **Name** field, enter an appropriate name for the image.
6. Optional: In the **Description** field, enter a description for the image.

7. Optional: From the **Accelerator identifier** list, select an identifier to set its accelerator as recommended with the image. If the image contains only one accelerator identifier, the identifier name displays by default.
8. Optional: Add software to the image. After the import has completed, the software is added to the image's meta-data and displayed on the workbench creation page.
 - a. Click the **Software** tab.
 - b. Click the **Add software** button.
 - c. Click **Edit** ().
 - d. Enter the **Software** name.
 - e. Enter the software **Version**.
 - f. Click **Confirm** () to confirm your entry.
 - g. To add additional software, click **Add software**, complete the relevant fields, and confirm your entry.
9. Optional: Add packages to the workbench images. After the import has completed, the packages are added to the image's meta-data and displayed on the workbench creation page.
 - a. Click the **Packages** tab.
 - b. Click the **Add package** button.
 - c. Click **Edit** ().
 - d. Enter the **Package** name.
 - e. Enter the package **Version**. For example, type **3.16.7**.
 - f. Click **Confirm** () to confirm your entry.
 - g. To add an additional package, click **Add package**, complete the relevant fields, and confirm your entry.
10. Click **Import**.

Verification

- The image that you imported is displayed in the table on the **Workbench images** page.
- Your custom image is available for selection when a user creates a workbench.

Additional resources

- [Managing image streams](#)
- [Understanding build configurations](#)

CHAPTER 4. INTRODUCING THE KUBERNETES GATEWAY API FOR CUSTOM IMAGE MIGRATION

Previously, the standard setup of JupyterLab and code-server leveraged OpenShift Routes and unique hostnames or subdomains for each workbench. Traffic was directed to the pod, and the application inside the pod usually served content from the root path (/). Red Hat OpenShift AI now leverages the Kubernetes Gateway API for request routing. This architecture replaces individual OpenShift Routes and Ingress resources with path-based routing through a single entry point.

4.1. UNDERSTANDING PATH-BASED ROUTING

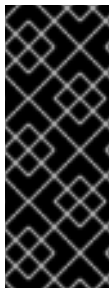
Path-based routing is a traffic management strategy where an API Gateway or load balancer sends requests to specific backend services based on the URL path provided by the client.

The Kubernetes Gateway API introduces path-based routing to your workbench image updating workflow.

Key architectural change

- OpenShift Route (Previous)
External: /notebook/user/workbench/app/ → Route strips prefix → Container receives: /app/
- Gateway API (New)
External: /notebook/user/workbench/app/ → Gateway preserves full path (path-based routing) → Container receives: /notebook/user/workbench/app/

If your application redirects to paths outside of the `${NB_PREFIX}`, the Gateway cannot route those requests back to your application. The path-based matching at the Gateway level requires all traffic to stay within the configured prefix.



IMPORTANT

Your application (or reverse proxy) must handle the full path including the prefix and never redirect outside of it.

Any request from a browser to a different path will **not** be routed to the workbench container. This is because the routing, handled by the Gateway API uses the same value as the environment variable (`NB_PREFIX`) that is injected into the workbench at runtime.

4.2. MIGRATING YOUR CUSTOM WORKBENCH IMAGES TO THE KUBERNETES GATEWAY API

To ensure that your workbench maintains compatibility with the latest versions of Red Hat OpenShift AI, update your custom workbench images to the Kubernetes Gateway API to support this path-based model.

Prerequisites

- You are a Red Hat OpenShift AI user with OpenShift AI administrator privileges.
- Your OpenShift cluster version is at least version 4.19.
- Your Red Hat OpenShift AI Operator version is at least version 3.0.

- Your system can build custom workbench images

Procedure

1. Implement required health check endpoints. The Gateway API requires a specific endpoint to verify workbench pod availability.
 - Configure the application to return an **HTTP 200** status at: **GET /{NB_PREFIX}/api**.
2. Configure culler support:
To support automatic idleness culling, the workbench must expose kernels and terminal status as JSON arrays. Ensure the application implements these endpoints:
 - **GET /{NB_PREFIX}/api/kernels**
 - **GET /{NB_PREFIX}/api/terminals**
3. Use relative URLs throughout your application.



NOTE

Avoid hardcoded absolute paths such as **/static/app.js** or **/api/data**. These paths bypass the **{NB_PREFIX}** and cannot be routed by the Gateway. Instead, use relative paths like **static/app.js** or **api/data**, which resolve correctly within the prefixed context.

If your application contains hardcoded absolute URLs that cannot be modified, you must use NGINX as a reverse proxy to handle path translation. See *Migrating your custom workbench images to the Kubernetes Gateway API using NGINX* for configuration details.

4. Configure your application's base path.
If your framework supports a configurable base path or root path, set it to the value of the **NB_PREFIX** environment variable. This ensures that generated URLs, redirects, and static asset references include the correct prefix.

For example, in Python frameworks you can set **root_path** (FastAPI) or **APPLICATION_ROOT** (Flask). In JavaScript frameworks, configure the base URL or public path setting accordingly.

Verification

- Click on your workbench through the **Workbenches** tab on the project details page. Verify that the page loads correctly. If the page displays without formatting or functional errors, the **{NB_PREFIX}** variable is configured correctly.
- Use the OpenShift CLI to verify the gateway can reach the health endpoint:

```
oc exec <pod-name> -- curl -I http://localhost:8888/${NB_PREFIX}/api
```

The command should return an **HTTP/1.1 200 OK** response.

- Monitor the Gateway Controller logs for **No route matched** errors. These errors indicate that the workbench is sending responses that lack the required path-based routing prefix.

4.3. MIGRATING YOUR CUSTOM WORKBENCH IMAGES TO THE KUBERNETES GATEWAY API USING NGINX

If your application does not support base path configuration, you will need NGINX to handle the path translation.

Prerequisites

- You are a Red Hat OpenShift AI user with OpenShift AI administrator privileges.
- Your OpenShift cluster version is at least version 4.19.
- Your Red Hat OpenShift AI Operator version is at least version 3.0.
- Your system can build custom workbench images.

Procedure

1. Update redirects to preserve **NB_PREFIX**. All redirects must include **`\${NB\_PREFIX}`** to keep requests within the Gateway route:

```
# BAD - Strips prefix
location = `${NB_PREFIX}` {
    return 302 $custom_scheme://$http_host/myapp/;
}

# GOOD - Preserves prefix
location `${NB_PREFIX}` {
    return 302 $custom_scheme://$http_host`${NB_PREFIX}`/myapp/;
}
```

2. Add a prefix-aware proxy location that matches the full prefixed path, and strips the prefix before proxying.

```
location `${NB_PREFIX}`/myapp/ {
    # Strip the prefix before proxying to backend
    rewrite ^`${NB_PREFIX}`/myapp/(.*)$ /$1 break;

    # Proxy to your application
    proxy_pass http://localhost:8080/;
    proxy_http_version 1.1;

    # Essential for WebSocket support
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection $connection_upgrade;

    # Long timeout for interactive sessions
    proxy_read_timeout 20d;

    # Pass through important headers
    proxy_set_header Host $http_host;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $custom_scheme;
}
```

3. Update health check endpoints to preserve the prefix:

```
# Health check endpoint
location = ${NB_PREFIX}/api {
    return 302 ${NB_PREFIX}/myapp/healthz/;
    access_log off;
}
```

Verification

- Verify that the page loads correctly. If the page displays without formatting or functional errors, the `${NB_PREFIX}` variable is configured correctly.
- Use the OpenShift CLI to verify the gateway can reach the health endpoint:

```
oc exec <pod-name> -- curl -I http://localhost:8888/${NB_PREFIX}/api
```

The command should return an **HTTP/1.1 200 OK** response.

- Monitor the Gateway Controller logs for **No route matched** errors. These errors indicate that the workbench is sending responses that lack the required path-based routing prefix.

CHAPTER 5. MANAGING CLUSTER PVC SIZE

5.1. CONFIGURING THE DEFAULT PVC SIZE FOR YOUR CLUSTER

To configure how resources are claimed within your OpenShift AI cluster, you can change the default size of the cluster's persistent volume claim (PVC) ensuring that the storage requested matches your common storage workflow. PVCs are requests for resources in your cluster and also act as claim checks to the resource.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.



NOTE

Changing the PVC setting restarts the workbench pod and makes it unavailable for up to 30 seconds. As a workaround, it is recommended that you perform this action outside of your organization's typical working day.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings**.
2. Under **PVC size**, enter a new size in gibibytes or mebibytes.
3. Click **Save changes**.

Verification

- New PVCs are created with the default storage size that you configured.

Additional resources

- [Understanding persistent storage](#)

5.2. RESTORING THE DEFAULT PVC SIZE FOR YOUR CLUSTER

To change the size of resources utilized within your OpenShift AI cluster, you can restore the default size of your cluster's persistent volume claim (PVC).

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings**.
2. Click **Restore Default** to restore the default PVC size of 20GiB.
3. Click **Save changes**.

Verification

- New PVCs are created with the default storage size of 20 GiB.

Additional resources

- [Understanding persistent storage](#)

CHAPTER 6. MANAGING CONNECTION TYPES

In Red Hat OpenShift AI, a connection comprises environment variables along with their respective values. Data scientists can add connections to project resources, such as workbenches and model servers.

When a data scientist creates a connection, they start by selecting a connection type. Connection types are templates that include customizable fields and optional default values. Starting with a connection type decreases the time required by a user to add connections to data sources and sinks. OpenShift AI includes pre-installed connection types for S3-compatible object storage databases and URI-based repositories.

As an OpenShift AI administrator, you can manage connection types for users in your organization as follows:

- View connection types and preview user connection forms
- Create a connection type
- Duplicate an existing connection type
- Edit a connection type
- Delete a custom connection type
- Enable or disable a connection type in a project, to control whether it is available as an option to users when they create a connection

6.1. VIEWING CONNECTION TYPES

As an OpenShift AI administrator, you can view the connection types that are available in a project.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Environment setup** → **Connection types**. The **Connection types** page opens, displaying the available connection types for the current project.
2. Optionally, you can select the Options menu  and then click **Preview** to see how the connection form associated with the connection type is displayed to your users.

6.2. CREATING A CONNECTION TYPE

As an OpenShift AI administrator, you can create a connection type for users in your organization.

You can create a new connection type as described in this procedure or you can create a copy of an existing connection type and edit it, as described in [Duplicating a connection type](#).

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- You know the environment variables that are required or optional for the connection type that you want to create.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Environment setup → Connection types**. The **Connection types** page opens, displaying the available connection types.
2. Click **Create connection type**
3. In the **Create connection type** form, enter the following information:
 - a. Enter a name for the connection type.
A resource name is generated based on the name of the connection type. A resource name is the label for the underlying resource in OpenShift.
 - b. Optionally, edit the default resource name. Note that you cannot change the resource name after you create the connection type.
 - c. Optionally, provide a description of the connection type.
 - d. Specify at least one category label. By default, the category labels are database, model registry, object storage, and URI. Optionally, you can create a new category by typing the new category label in the field. You can specify more than one category.
The category label is for descriptive purposes only. It allows you and the users in your organization to sort the available connection types when viewing them in the OpenShift AI dashboard interface.
 - e. Check the **Enable users in your organization to use this connection type when adding connections** option if you want the connection type to appear in the list of connections available to users, for example, when they configure a workbench, a model server, or a pipeline.
Note that you can also enable/disable the connection type after you create it.
 - f. For the **Fields** section, add the fields and section headings that you want your users to see in the form when they add a connection to a project resource (such as a workbench or a model server).
Note that the connection name and description fields are included by default, so you do not need to add them.
 - Optionally, select a model serving compatible type to automatically add the fields required to use its corresponding model serving method.
 - Click **Add field** to add a field to prompt users to input information, and optionally assign default values to those fields.
 - Click **Add section heading** to organize the fields under headings.
4. Click **Preview** to open a preview of the connection form as it will appear to your users.
5. Click **Save**.

Verification

1. On the **Settings → Environment setup → Connection types** page, the new connection type is displayed in the list.

6.3. DUPLICATING A CONNECTION TYPE

As an OpenShift AI administrator, you can create a new connection type by duplicating an existing one, as described in this procedure, or you can create a new connection type as described in [Creating a connection type](#).

You might also want to duplicate a connection type if you want to create versions of a specific connection type.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Environment setup → Connection types**.
2. From the list of available connection types, find the connection type that you want to duplicate.

Optionally, you can select the Options menu  and then click **Preview** to see how the related connection form is displayed to your users.

3. Click the Options menu , and then click **Duplicate**.
The **Create connection type** form is displayed populated with the information from the connection type that you duplicated.
4. Edit the form according to your use case.
5. Click **Preview** to open a preview of the connection form as it will appear to your users and verify that the form is displayed as you expect.
6. Click **Save**.

Verification

In the **Settings → Environment setup → Connection types** page, the duplicated connection type is displayed in the list.

6.4. EDITING A CONNECTION TYPE

As an OpenShift AI administrator, you can edit a connection type for users in your organization.

Note that you cannot edit the connection types that are pre-installed with OpenShift AI. Instead, you have the option of duplicating a pre-installed connection type, as described in [Duplicating a connection type](#).

When you edit a connection type, your edits do not apply to any existing connections that users previously created. If you want to keep track of previous versions of this connection type, consider duplicating it instead of editing it.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The connection type must exist and must not be a pre-installed connection type (which you are unable to edit).

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Environment setup** → **Connection types**.
2. From the list of available connection types, find the connection type that you want to edit.
3. Click the Options menu , and then click **Edit**.
The **Edit connection type** form is displayed.
4. Edit the form fields and sections.
5. Click **Preview** to open a preview of the connection form as it will appear to your users and verify that the form is displayed as you expect.
6. Click **Save**.

Verification

In the **Settings** → **Environment setup** → **Connection types** page, the duplicated connection type is displayed in the list.

6.5. ENABLING A CONNECTION TYPE

As an OpenShift AI administrator, you can enable or disable a connection type to control whether it is available as an option to your users when they create a connection.

Note that if you disable a connection type, any existing connections that your users created based on that connection type are not effected.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The connection type that you want to enable exists in your project, either pre-installed or created by a user with administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Environment setup** → **Connection types**.
2. From the list of available connection types, find the connection type that you want to enable or disable.
3. On the row containing the connection type, click the toggle in the **Enable** column.

Verification

- If you enabled a connection type, it is available for selection when a user adds a connection to a project resource (for example, a workbench or model server).

- If you disabled a connection type, it does not show in the list of available connection types when a user adds a connection to a project resource.

6.6. DELETING A CONNECTION TYPE

As an OpenShift AI administrator, you can delete a connection type that you or another administrator created.

Note that you cannot delete the connection types that are pre-installed with OpenShift AI. Instead, you have the option of disabling them so that they are not visible to your users, as described in [Enabling a connection type](#).

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The connection type must exist and must not be a pre-installed connection type (which you are unable to delete).

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Environment setup** → **Connection types**.
2. From the list of available connection types, find the connection type that you want to delete.

Optionally, you can select the Options menu  and then click **Preview** to see how the related connection form is displayed to your users.

3. Click the Options menu , and then click **Delete**.
4. In the **Delete connection type?** form, type the name of the connection type that you want to delete and then click **Delete**.

Verification

In the **Settings** → **Environment setup** → **Connection types** page, the connection type is no longer displayed in the list.

CHAPTER 7. MANAGING STORAGE CLASSES

OpenShift cluster administrators use storage classes to describe the different types of storage that is available in their cluster. These storage types can represent different quality-of-service levels, backup policies, or other custom policies set by cluster administrators.

7.1. ABOUT PERSISTENT STORAGE

OpenShift AI uses persistent storage to support workbenches, project data, and model training.

Persistent storage is provisioned through OpenShift storage classes and persistent volumes. Volume provisioning and data access are determined by access modes.

Understanding storage classes and access modes can help you choose the right storage for your use case and avoid potential risks when sharing data across multiple workbenches.

7.1.1. Storage classes in OpenShift AI

Storage classes in OpenShift AI are available from the underlying OpenShift cluster. A storage class defines how persistent volumes are provisioned, including which storage backend is used and what access modes the provisioned volumes can support. For more information, see [Dynamic provisioning](#) in the OpenShift documentation.

Cluster administrators create and configure storage classes in the OpenShift cluster. These storage classes provision persistent volumes that support one or more access modes, depending on the capabilities of the storage backend. OpenShift AI administrators then enable specific storage classes and access modes for use in OpenShift AI.

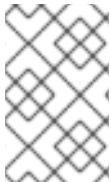
When adding cluster storage to your project or workbench, you can choose from any enabled storage classes and access modes.

7.1.2. Access modes

Storage classes create persistent volumes that can support different access modes, depending on the storage backend. Access modes control how a volume can be mounted and used by one or more workbenches. If a storage class allows more than one access mode, you can select the one that best fits your needs when you request storage. All persistent volumes support **ReadWriteOnce (RWO)** by default.

Access mode	Description
ReadWriteOnce (RWO) (Default)	The storage can be attached to a single workbench or pod at a time and is ideal for most individual workloads. RWO is always enabled by default and cannot be disabled by the administrator.
ReadWriteMany (RWX)	The storage can be attached to many workbenches simultaneously. RWX enables shared data access, but can introduce data risks.
ReadOnlyMany (ROX)	The storage can be attached to many workbenches as read-only. ROX is useful for sharing reference data without allowing changes.

Access mode	Description
ReadWriteOncePod (RWOP)	The storage can be attached to a single pod on a single node with read-write permissions. RWOP is similar to RWO but includes additional node-level restrictions.



NOTE

You can enable access modes that are required. A warning is displayed if you request an access mode with unknown support, but you can continue to select **Save** to create the storage class with the selected access mode.

7.1.2.1. Using shared storage (RWX)

The **ReadWriteMany (RWX)** access mode allows multiple workbenches to access and write to the same storage volume at the same time. Use **RWX** access mode for collaborative work where multiple users need to access shared datasets or project files.

However, shared storage introduces several risks:

- **Data corruption or data loss** If multiple workbenches modify the same part of a file simultaneously, the data can become corrupted or lost. Ensure your applications or workflows are designed to safely handle shared access, for example, by using file locking or database transactions.
- **Security and privacy.** If a workbench with access to shared storage is compromised, all data on that volume might be at risk. Only share sensitive data with trusted workbenches and users.

To use shared storage safely:

- Ensure that your tools or workflows are designed to work with shared storage and can manage simultaneous writes. For example, use databases or distributed data processing frameworks.
- Be cautious with changes. Deleting or editing files affects everyone who shares the volume.
- Back up your data regularly, which can help prevent data loss due to mistakes or misconfigurations.
- Limit access to RWX volumes to trusted users and secure workbenches.
- Use **ReadWriteMany (RWX)** only when collaboration on a shared volume is required. For most individual tasks, **ReadWriteOnce (RWO)** is ideal because only one workbench can write to the volume at a time.

7.2. CONFIGURING STORAGE CLASS SETTINGS

As an OpenShift AI administrator, you can manage the following OpenShift cluster storage class settings for use within OpenShift AI:

- Display name

- Description
- Access modes
- Whether users can use the storage class when creating or editing cluster storage

These settings do not impact the storage class within OpenShift.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings** → **Storage classes**.
The **Storage classes** page opens, displaying the storage classes for your cluster as defined in OpenShift.
2. To enable or disable a storage class for users, on the row containing the storage class, click the toggle in the **Enable** column.
3. To edit a storage class, on the row containing the storage class, click the action menu (**:**) and then select **Edit**.
The **Edit storage class details** dialog opens.
4. Optional: In the **Display Name** field, update the name for the storage class. This name is used only in OpenShift AI and does not impact the storage class within OpenShift.
5. Optional: In the **Description** field, update the description for the storage class. This description is used only in OpenShift AI and does not impact the storage class within OpenShift.
6. For storage classes that support multiple access modes, select an **Access mode** to define how the volume can be accessed. For more information, see [About persistent storage](#).



NOTE

A warning is displayed if you request an access mode with unknown support, but you can continue to select **Save** to create the storage class with the selected access mode. Only the access modes that have been enabled for the storage class by cluster and OpenShift AI administrators are visible.

7. Click **Save**.

Verification

- If you enabled a storage class, the storage class is available for selection when a user adds cluster storage to a project or workbench.
- If you disabled a storage class, the storage class is not available for selection when a user adds cluster storage to a project or workbench.
- If you edited a storage class name, the updated storage class name is displayed when a user adds cluster storage to a project or workbench.

Additional resources

- [Storage classes in OpenShift](#)

7.3. CONFIGURING THE DEFAULT STORAGE CLASS FOR YOUR CLUSTER

As an OpenShift AI administrator, you can configure the default storage class for OpenShift AI to be different from the default storage class in OpenShift.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings** → **Storage classes**. The **Storage classes** page opens, displaying the storage classes for your cluster as defined in OpenShift.
2. If the storage class that you want to set as the default is not enabled, on the row containing the storage class, click the toggle in the **Enable** column.
3. To set a storage class as the default for OpenShift AI, on the row containing the storage class, select **Set as default**

Verification

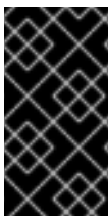
- When a user adds cluster storage to a project or workbench, the default storage class that you configured is automatically selected.

Additional resources

- [Storage classes in OpenShift](#)

7.4. OVERVIEW OF OBJECT STORAGE ENDPOINTS

To ensure correct configuration of object storage in OpenShift AI, you must format endpoints correctly for the different types of object storage supported. These instructions are for formatting endpoints for Amazon S3, MinIO, or other S3-compatible storage solutions, minimizing configuration errors and ensuring compatibility.



IMPORTANT

Properly formatted endpoints enable connectivity and reduce the risk of misconfigurations. Use the appropriate endpoint format for your object storage type. Improper formatting might cause connection errors or restrict access to storage resources.

7.4.1. MinIO (On-Cluster)

For on-cluster MinIO instances, use a local endpoint URL format. Ensure the following when configuring MinIO endpoints:

- Prefix the endpoint with **http://** or **https://** depending on your MinIO security setup.

- Include the cluster IP or hostname, followed by the port number if specified.
- Use a port number if your MinIO instance requires one (default is typically **9000**).

Example:

```
http://minio-cluster.local:9000
```



NOTE

Verify that the MinIO instance is accessible within the cluster by checking your cluster DNS settings and network configurations.

7.4.2. Amazon S3

When configuring endpoints for Amazon S3, use region-specific URLs. Amazon S3 endpoints generally follow this format:

- Prefix the endpoint with **https://**.
- Format as **<bucket-name>.s3.<region>.amazonaws.com**, where **<bucket-name>** is the name of your S3 bucket, and **<region>** is the AWS region code (for example, **us-west-1**, **eu-central-1**).

Example:

```
https://my-bucket.s3.us-west-2.amazonaws.com
```



NOTE

For improved security and compliance, ensure that your Amazon S3 bucket is in the correct region.

7.4.3. Other S3-Compatible Object Stores

For S3-compatible storage solutions other than Amazon S3, follow the specific endpoint format required by your provider. Generally, these endpoints include the following items:

- The provider base URL, prefixed with **https://**.
- The bucket name and region parameters as specified by the provider.
- Review the documentation from your S3-compatible provider to confirm required endpoint formats.
- Replace placeholder values like **<bucket-name>** and **<region>** with your specific configuration details.

**WARNING**

Incorrectly formatted endpoints for S3-compatible providers might lead to access denial. Always verify the format in your storage provider documentation to ensure compatibility.

7.4.4. Verification and Troubleshooting

After configuring endpoints, verify connectivity by performing a test upload or accessing the object storage directly through the OpenShift AI dashboard. For troubleshooting, check the following items:

- **Network Accessibility:** Confirm that the endpoint is reachable from your OpenShift AI cluster.
- **Authentication:** Ensure correct access credentials for each storage type.
- **Endpoint Accuracy:** Double-check the endpoint URL format for any typos or missing components.

Additional resources

- Amazon S3 Region and Endpoint Documentation: [AWS S3 Documentation](#)

CHAPTER 8. MANAGING BASIC WORKBENCHES

8.1. ACCESSING THE ADMINISTRATION INTERFACE FOR BASIC WORKBENCHES

You can use the administration interface to control basic workbenches in your Red Hat OpenShift AI environment.

Prerequisite

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

To access the administration interface for basic workbenches from OpenShift AI, perform the following actions:

1. In OpenShift AI, in the **Applications** section of the left menu, click **Enabled**.
2. Locate the **Start basic workbench** tile and click **Open application**.
3. On the page that opens, click the **Administration** tab.
The **Administration** page opens.

Verification

- You can see the administration interface for basic workbenches.

8.2. STARTING BASIC WORKBENCHES OWNED BY OTHER USERS

OpenShift AI administrators can start a basic workbench for another existing user from the administration interface for basic workbenches.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- You have launched the **Start basic workbench** application, as described in [Starting a basic workbench](#).

Procedure

1. On the page that opens when you launch a basic workbench, click the **Administration** tab.
2. On the **Administration** tab, perform the following actions:
 - a. In the **Users** section, locate the user whose workbench you want to start.
 - b. Click **Start workbench** beside the relevant user.
 - c. Complete the **Start a basic workbench** page.
 - d. Optional: Select the **Start workbench in current tab** checkbox if necessary.

e. Click **Start workbench**

After the server starts, you see one of the following behaviors:

- If you previously selected the **Start workbench in current tab** checkbox, the JupyterLab interface opens in the current tab of your web browser.
- If you did not previously select the **Start workbench in current tab** checkbox, the **Workbench status** dialog box prompts you to open the server in a new browser tab or in the current tab.

The JupyterLab interface opens according to your selection.

Verification

- The JupyterLab interface opens.

8.3. ACCESSING BASIC WORKBENCHES OWNED BY OTHER USERS

OpenShift AI administrators can access basic workbenches that are owned by other users to correct configuration errors or to help them troubleshoot problems with their environment.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- You have launched the **Start basic workbench** application, as described in [Starting a basic workbench](#).
- The workbench that you want to access is running.

Procedure

1. On the page that opens when you launch a basic workbench, click the **Administration** tab.
2. On the **Administration** page, perform the following actions:
 - a. In the **Users** section, locate the user that the workbench belongs to.
 - b. Click **View server** beside the relevant user.
 - c. On the **Workbench control panel** page, click **Access workbench**.

Verification

- The JupyterLab interface opens in the user's workbench.

8.4. STOPPING BASIC WORKBENCHES OWNED BY OTHER USERS

OpenShift AI administrators can stop basic workbenches that are owned by other users to reduce resource consumption on the cluster, or as part of removing a user and their resources from the cluster.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

- You have launched the **Start basic workbench** application, as described in [Starting a basic workbench](#).
- The workbench that you want to stop is running.

Procedure

1. On the page that opens when you launch a basic workbench, click the **Administration** tab.
2. Stop one or more servers.
 - If you want to stop one or more specific servers, perform the following actions:
 - i. In the **Users** section, locate the user that the workbench belongs to.
 - ii. To stop the workbench, perform one of the following actions:
 - Click the action menu (**:**) beside the relevant user and select **Stop server**.
 - Click **View server** beside the relevant user and then click **Stop workbench**. The **Stop server** dialog box opens.
 - iii. Click **Stop server**.
 - If you want to stop all workbenches, perform the following actions:
 - i. Click the **Stop all workbenches** button.
 - ii. Click **OK** to confirm stopping all servers.

Verification

- The **Stop server** link beside each server changes to a **Start workbench** link when the workbench has stopped.

8.5. STOPPING IDLE WORKBENCHES

You can reduce resource usage in your OpenShift AI deployment by stopping workbenches that have been idle (without logged in users) for a period of time. This is useful when resource demand in the cluster is high. By default, idle workbenches are not stopped after a specific time limit.



NOTE

If you have configured your cluster settings to disconnect all users from a cluster after a specified time limit, then this setting takes precedence over the idle workbench time limit. Users are logged out of the cluster when their session duration reaches the cluster-wide time limit.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings**.

2. Under **Idle workbench timeout**, select **Stop idle workbenches after defined period**.
3. Enter a time limit, in **hours** and **minutes**, for when idle workbenches are stopped.
4. Click **Save changes**.

Verification

- In OpenShift, go to **Workloads → ConfigMaps** and open the **notebook-controller-culler-config** ConfigMap in the **redhat-ods-applications** project to verify that it contains the following culling configuration settings:
 - **ENABLE_CULLING**: Specifies if the culling feature is enabled or disabled (this is **false** by default).
 - **IDLENESS_CHECK_PERIOD**: The polling frequency to check for a notebook's last known activity (in minutes).
 - **CULL_IDLE_TIME**: The maximum allotted time to scale an inactive notebook to zero (in minutes).
- Idle workbenches stop at the time limit that you set.

8.6. ADDING WORKBENCH POD TOLERATIONS

If you want to dedicate certain machine pools to only running workbench pods, you can allow workbench pods to be scheduled on specific nodes by adding a toleration. Taints and tolerations allow a node to control which pods should (or should not) be scheduled on them. For more information, see [Understanding taints and tolerations](#).

This capability is useful if you want to make sure that workbenches are placed on nodes that can handle their needs. By preventing other workloads from running on these specific nodes, you can ensure that the necessary resources are available to users who need to work with large workbench sizes.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- You are familiar with OpenShift taints and tolerations, as described in [Understanding taints and tolerations](#).

Procedure

1. From the OpenShift AI dashboard, click **Settings → Cluster settings**.
2. Under **Workbench pod tolerations**, select **Add a toleration to workbench pods to allow them to be scheduled to tainted nodes**.
3. In the **Toleration key for workbench pods** field, enter a toleration key. The key is any string, up to 253 characters. The key must begin with a letter or number, and can contain letters, numbers, hyphens, dots, and underscores. For example, **workbenches-only**.
4. Click **Save changes**. The toleration key is applied to new workbench pods when they are created.

For existing workbench pods, the toleration key is applied when the workbench pods are restarted.

If you are using a basic workbench, see [Updating workbench settings by restarting your workbench](#) . If you are using a workbench in a project, see [Starting a workbench](#) .

Next step

In OpenShift, add a matching taint key (with any value) to the machine pools that you want to dedicate to workbenches. For more information, see [Controlling pod placement using node taints](#) .

For more information, see [Adding taints to a machine pool](#) .

Verification

1. In the OpenShift console, select your project, then click **Workloads** → **StatefulSet**. You can see how many pods are running, either zero or one, depending on whether your workbench is currently started or stopped.
2. Search for your workbench pod name and then click the name to open the pod details page.
3. Confirm that the assigned **Node** and **Tolerations** are correct.

8.7. TROUBLESHOOTING COMMON PROBLEMS IN WORKBENCHES FOR ADMINISTRATORS

If your users are experiencing errors in Red Hat OpenShift AI relating to Jupyter, their Jupyter notebooks, or their workbench, read this section to understand what could be causing the problem, and how to resolve the problem.

If you cannot see the problem here or in the release notes, contact Red Hat Support.

8.7.1. A user receives a 404: Page not found error when logging in to Jupyter

Problem

If you have configured OpenShift AI user groups, the user name might not be added to the default user group for OpenShift AI.

Diagnosis

Check whether the user is part of the default user group.

1. Find the names of groups allowed access to Jupyter.
 - a. Log in to the OpenShift web console.
 - b. Click **User Management** → **Groups**.
 - c. Click the name of your user group, for example, **rhods-users**.
The **Group details** page for that group is displayed.
2. Click the **Details** tab for the group and confirm that the **Users** section for the relevant group contains the users who have permission to access Jupyter.

Resolution

- If the user is not added to any of the groups with permission access to Jupyter, follow [Adding users to OpenShift AI user groups](#) to add them.
- If the user is already added to a group with permission to access Jupyter, contact Red Hat Support.

8.7.2. A user's workbench does not start

Problem

The OpenShift cluster that hosts the user's workbench might not have access to enough resources, or the workbench pod may have failed.

Diagnosis

1. Log in to the OpenShift web console.
2. Delete and restart the workbench pod for this user.
 - a. Click **Workloads** → **Pods** and set the **Project** to **rhods-notebooks** or your custom workbench namespace.
 - b. Search for the workbench pod that belongs to this user, for example, **jupyter-nb-`<username>-*`**.
If the workbench pod exists, an intermittent failure might have occurred in the workbench pod.

If the workbench pod for the user does not exist, continue with diagnosis.
3. Check the resources currently available in the OpenShift cluster against the resources required by the selected workbench image.
If worker nodes with sufficient CPU and RAM are available for scheduling in the cluster, continue with diagnosis.
4. Check the state of the workbench pod.

Resolution

- If there was an intermittent failure of the workbench pod:
 - a. Delete the workbench pod that belongs to the user.
 - b. Ask the user to start their workbench again.
- If the workbench does not have sufficient resources to run the selected workbench image, either add more resources to the OpenShift cluster, or choose a smaller image size.
- If the workbench pod is in a **FAILED** state:
 - a. Retrieve the logs for the **jupyter-nb-*** pod and send them to Red Hat Support for further evaluation.
 - b. Delete the **jupyter-nb-*** pod.
- If none of the previous resolutions apply, contact Red Hat Support.

8.7.3. The user receives a database or disk is full error or a no space left on device error when they run notebook cells

Problem

The user might have run out of storage space on their workbench.

Diagnosis

1. Log in to Jupyter and start the workbench that belongs to the user having problems. If the workbench does not start, follow these steps to check whether the user has run out of storage space:
 - a. Log in to the OpenShift web console.
 - b. Click **Workloads** → **Pods** and set the **Project** to **rhods-notebooks** or your custom workbench namespace.
 - c. Click the workbench pod that belongs to this user, for example, **jupyter-nb-`<idp>-<username>-*`**.
 - d. Click **Logs**. The user has exceeded their available capacity if you see lines similar to the following:

Unexpected error while saving file: XXXX database or disk is full

Resolution

- Increase the user's available storage by expanding their persistent volume: [Expanding persistent volumes](#)
- Work with the user to identify files that can be deleted from the **/opt/app-root/src** directory on their workbench to free up their existing storage space.



NOTE

When you delete files using the JupyterLab file explorer, the files move to the hidden **/opt/app-root/src/.local/share/Trash/files** folder in the persistent storage for the workbench. To free up storage space for workbenches, you must permanently delete these files.

CHAPTER 9. MANAGING THE COLLECTION OF USAGE DATA

Red Hat OpenShift AI administrators can choose whether to allow Red Hat to collect data about OpenShift AI usage in their cluster. Collecting this data allows Red Hat to monitor and improve our software and support. For further details about the data Red Hat collects, see [Usage data collection notice for OpenShift AI](#).

Usage data collection is enabled by default when you install OpenShift AI on your OpenShift cluster except when clusters are installed in a disconnected environment.

See [Disabling usage data collection](#) for instructions on disabling the collection of this data in your cluster. If you have disabled data collection on your cluster, and you want to enable it again, see [Enabling usage data collection](#) for more information.

9.1. USAGE DATA COLLECTION NOTICE FOR OPENSIFT AI

In connection with your use of this Red Hat offering, Red Hat may collect usage data about your use of the software. This data allows Red Hat to monitor the software and to improve Red Hat offerings and support, including identifying, troubleshooting, and responding to issues that impact users.

What information does Red Hat collect?

Tools within the software monitor various metrics and this information is transmitted to Red Hat. Metrics include information such as:

- Information about applications enabled in the product dashboard.
- The deployment sizes used (that is, the CPU and memory resources allocated).
- Information about documentation resources accessed from the product dashboard.
- The name of the notebook images used (that is, Minimal Python, Standard Data Science, and other images.).
- A unique random identifier that generates during the initial user login to associate data to a particular username.
- Usage information about components, features, and extensions.

Third Party Service Providers

Red Hat uses certain third party service providers to collect the telemetry data.

Security

Red Hat employs technical and organizational measures designed to protect the usage data.

Personal Data

Red Hat does not intend to collect personal information. If Red Hat discovers that personal information has been inadvertently received, Red Hat will delete such personal information and treat such personal information in accordance with Red Hat's Privacy Statement. For more information about Red Hat's privacy practices, see Red Hat's [Privacy Statement](#).

Enabling and Disabling Usage Data

You can disable or enable usage data by following the instructions in [Disabling usage data collection](#) or [Enabling usage data collection](#).

9.2. ENABLING USAGE DATA COLLECTION

Red Hat OpenShift AI administrators can select whether to allow Red Hat to collect data about OpenShift AI usage in their cluster. Usage data collection is enabled by default when you install OpenShift AI on your OpenShift cluster except when clusters are installed in a disconnected environment. If you have disabled data collection previously, you can re-enable it by following these steps.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings**.
2. Locate the **Usage data collection** section.
3. Select the **Allow collection of usage data** checkbox.
4. Click **Save changes**.

Verification

- A notification is shown when settings are updated: **Settings changes saved**.

Additional resources

- [Usage data collection notice for OpenShift AI](#)

9.3. DISABLING USAGE DATA COLLECTION

Red Hat OpenShift AI administrators can choose whether to allow Red Hat to collect data about OpenShift AI usage in their cluster. Usage data collection is enabled by default when you install OpenShift AI on your OpenShift cluster except when clusters are installed in a disconnected environment.

You can disable data collection by following these steps.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Cluster settings**.
2. Locate the **Usage data collection** section.
3. Clear the **Allow collection of usage data** checkbox.
4. Click **Save changes**.

Verification

- A notification is shown when settings are updated: **Settings changes saved**.

Additional resources

- [Usage data collection notice for OpenShift AI](#)